

Jichen YAO

Email: yaojc825@163.com | Mobile: +86 13591179625
195 Chuangxin Road, Baita Street, Hunnan District, Shenyang, Liaoning Province, China

EDUCATION

Northeastern University

Bachelor's Degree

Sep 2023 – Jul 2027 (Expected)

Liaoning, China

- Major: Computer Science and Technology | GPA: 3.90/5.0
- Core Courses: Fundamentals of Programming (C Language), Object-Oriented Programming (C++), Intelligent Electronics and Information Technology, Algorithm Design and Analysis, Numerical Analysis, Digital Logic and Digital Systems, Basic Training of Information Technology, Data Structures, Probability Theory and Mathematical Statistics, Linear Algebra, Advanced Mathematics, Discrete Mathematics

RESEARCH PROJECTS

Fractal Spring Pyramid Memory Architecture for Long Context in LLMs

Mar 2026 – Present

Advisor: Assoc. Prof. Shi Feng

- Proposed a Fractal Spring Architecture with a three-level hierarchical memory (Micro/Meso/Macro Coils) to compress KV Cache from linear to logarithmic growth.
- Implemented an Adaptive Feature-level Gate based on ReZero mechanism, enabling the model to dynamically decide memory injection amounts without disrupting local semantics.
- Developed the entire training pipeline, including memory distillation and diversity loss optimization. Achieved promising results in Multi-passkey tasks, demonstrating robust retrieval capabilities under extreme noise.

Cognitive-Inspired Strategy Optimization for Emotional Support Dialogue (ESC)

Dec 2025 – May 2026

Advisor: Assoc. Prof. Shi Feng

- Developed a cognitive constraint-driven preference optimization pipeline, abstracting 5 psychological principles into multi-stage prompts to guide strategy alignment and mitigate premature problem-solving.
- Redesigned the MCTS framework's evaluation mechanisms by introducing a Cognitive Alignment reward signal to penalize premature advice and reinforce validation-before-intervention pacing.
- Generated **10k+** dialogue preference pairs through an MCTS rollout-to-preference extraction pipeline. Conducted comparative ablation studies and fine-tuned LLaMA models via LoRA and DPO using the LLaMA-Factory framework.
- Designed a robust evaluation framework (point-wise, pairwise blind, statistical tests) to analyze automated metric instability. Identified and analyzed the Verbosity Tax, validating it through cross-model contrast (LLaMA vs. Qwen).

Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification

Jul 2025 – Nov 2025

Advisor: Assoc. Prof. Shi Feng

- Proposed AMCA, an end-to-end model leveraging learnable aspect embeddings and multi-head cross-attention to extract semantics dynamically without external NLP tools.
- Redefined ABSA task objectives by shifting the focus from marginal classification accuracy to aspect detection completeness.
- Achieved state-of-the-art performance in aspect detection tasks across multiple datasets, demonstrating significant advantages in identifying implicit aspects through independent model design and systematic empirical validation.
- **Publication:** Jichen Yao, Shi Feng. "Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification." *Accepted by ICNC-FSKD 2026.*

EXTRACURRICULAR ACTIVITY

Dalian Municipal Culture and Tourism Bureau Warm Winter Volunteer Service

Jan 2025

Role: Volunteer | Organizer: Dalian Municipal Culture and Tourism Bureau

- Provided on-site assistance and information services to returning travelers at major transportation hubs including airports and railway stations.
- Delivered customer-oriented solutions to address inquiries regarding transportation, luggage, and local tourism information.
- Collaborated with a team of volunteers to ensure smooth operation of winter travel assistance initiative.

TECHNICAL SKILLS

- Languages: Mandarin (Native), English (TOEFL iBT Home Edition: 105)
- Programming: Proficient in Python, C, C++; Familiar with Shell scripting
- AI/ML Stack: PyTorch, LLaMA-Factory, HuggingFace, MCTS, DPO, Parameter-Efficient Fine-Tuning (LoRA)
- Tools: Git, Docker, LaTeX, Visual Studio Code, Linux Environment